Technical Supplement: Activation-Aligned Asynchronous Residual MPC for Position-Controlled Manipulators with Learned Dynamics

Public evidence bundle for ROBIO 2026

2026-07-29

1 Purpose and evidence boundary

This supplement provides the implementation, configuration, and diagnostic detail that is intentionally compressed in the conference manuscript. It is not a second experimental study and introduces no results beyond the frozen public evidence bundle. The source files named below are included in this same directory and are listed with SHA-256 digests in `PUBLIC_MANIFEST.json`.

All results use MuJoCo position-controlled robot models. ABB IRB 2400 is the primary robot; UR5e is an independently trained and calibrated nominal within-robot replication. The evidence does not establish physical-robot performance, hardware torque feasibility for ABB, hard-real-time scheduling, robust-MPC guarantees, shared-model transfer, or arbitrary-morphology generalization. The post-freeze effort sweep is diagnostic and did not select or replace the frozen primary configuration.

The repository also implements an optional budgeted uncertainty supervisor for threaded execution. Model A alone optimizes CEM; compatible GRU replicas then evaluate only the selected short-horizon trajectory, and normalized ensemble disagreement can drive hysteretic residual limiting or a strict zero-residual fallback to the projected current IK nominal. The supervisor was disabled in all frozen experiments, so this supplement reports its implementation boundary but no uncertainty-detection or safety-performance result.

2 Controller and execution configuration

The controller uses a one-layer GRU with hidden size 256 and history length 16. It predicts the velocity increment from aligned tokens $[q_t, \dot{q}_t, q_t^{\mathrm{ref}}]$ and is trained with one-step increment loss plus 20-step rollout loss on a disjoint 90/10 split. ABB and UR5e use separate XML models, data distributions, checkpoints, normalizers, task references, limits, and delay calibrations. The complete machine-readable contracts are `abb/manifests/control_manifest.json` and `ur5e/manifests/paper.json`.

Table 1: Frozen ABB controller settings. Per-joint vectors and all robot-specific artifacts are in the manifests.

Setting	Value	Setting	Value
Control interval	10 ms	History length	16
Horizon / samples / iterations	20 / 128 / 2	Elite ratio	0.08
Replan interval / delay D	5 / 6 steps	Guard	5 ms
Temporal discount	0.95	Projection	two-stage compiled
$w_q/w_{\dot{q}}/w_r/w_s$	1 / .10 / .20 / .05	$w_{\dot{r}}/w_{\ddot{r}}/w_{\mathrm{first}}$	.05 / .02 / .20
Joint / velocity barrier	10 / 5	Barrier max multiplier	2
Task position/orientation	1 / .25	Task scales	.05 m / .0873 rad
Feedback $K_q/K_{\dot{q}}/\delta r_{\max}$	.3 / .015 / .015 rad	Residual maximum	(0.12,.10,.12,.15,.15,.20) rad

Residual packets encode corrections, not reusable absolute commands. At packet age a, the full protocol reconstructs and projects

$$q_t^{\mathrm{cmd}} = \Pi_{v,a,\mathcal{Q}}(\bar{q}_t^{\mathrm{IK}} + r_a + \delta r_t; q_{t-1}^{\mathrm{cmd}}, \dot{q}_{t-1}^{\mathrm{cmd}}).$$

The launch-anchor replay counterfactual instead projects the cached planned absolute command q_a^{plan}. Its large failure is therefore evidence for the coordinate transformation required by an independently evolving nominal, not a competitive controller comparison. The implementation is in `mpc/delay_aware_runner.py`; the protocol definitions are in `mpc/delay_protocol.py`.

3 Artifacts, references, and calibration

The public manifests record XML/source hashes, TCP site and orientation, home/departure configurations, joint and actuator ranges, command limits, checkpoint and normalizer hashes, training-dataset and reference hashes, and the calibrated activation delay. The ABB delay is $D = 6$ and the UR5e delay is $D = 7$, each defined as

$$\left\lceil \left(P_{95}(t_{\text{snapshot}\to\text{publication}}) + 5\ \text{ms}\right)/10\ \text{ms}\right\rceil .$$

The ABB manifest records the checkpoint and normalizer SHA-256 digests; the corresponding UR5e values are recorded in `ur5e/manifests/paper.json`. Artifacts themselves are deliberately not committed to avoid distributing large learned-model files through Git.

4 Full diagnostics

Table 2: Held-out open-loop model errors pooled over steps 1:h. Errors are $q/\dot{q}$ RMSE in rad/rad s^{-1}; Div. is the divergent-rollout rate at $\sigma_u = 0.8$.

Robot, h	q RMSE, $\sigma_u = .5$	$\dot{q}$ RMSE, $\sigma_u = .5$	q RMSE, $\sigma_u = .8$	$\dot{q}$ RMSE, $\sigma_u = .8$	Div., $\sigma_u = .8$
ABB, 1	.00337	.55842	.00715	1.13450	0%
ABB, 20	.02164	.26490	.04411	.49777	2%
UR5e, 1	.00127	.12659	.00138	.13613	0%
UR5e, 20	.00956	.06916	.01057	.07509	0%

Table 3: ABB retained-candidate replay diagnostics. Spearman is computed within snapshots with defined ranks; all intervals are hierarchical bootstrap 95% intervals.

Metric	Estimate	95% interval
Within-snapshot Spearman	0.976	[0.917, 1.000]
Pairwise concordance	0.988	[0.957, 1.000]
Relative realized regret	0.192%	[0, 0.481]%
20-step branch q RMSE	0.889 mrad	[0.705, 1.121] mrad

Model validation and CEM ranking. Tables 2–3 provide the principal held-out and replay diagnostics without requiring the reader to inspect raw files. Full per-joint, excitation, and rollout-level diagnostics are in `abb/model_validation/gru_validation_aggregate.json` and the associated CSV files. Counterfactual candidate replay is in `abb/model_validation/formal_mpc_replay_aggregate.csv` and `claim_evidence/summaries/candidate_ranking.json`. It covers retained, projection-consistent online candidates only; its lower branch error must not be compared directly with the broader random-excitation validation.

Robustness and timing. ABB robustness applies L3/L6 single-factor payload, actuator-gain, TCP-force, and observation-noise perturbations. The full per-condition and cluster bootstrap results are in `robustness_and_timing` and `abb/summaries`. Wall-clock timing, packet behavior, projection activity, and the threaded-versus-virtual contrast are stored alongside them. All primary experiments had no planner failure or imposed joint, command-velocity, or command-acceleration violation; this does not imply a physical actuator limit.

Mechanism isolation. The frozen 20-case ABB matrix gives +2.82 mm for removing future alignment (95% CI $[1.27, 4.39]$ mm), +654.60 mm for launch-anchor absolute replay ($[528.68, 778.68]$ mm), and −0.95 mm for removing fast feedback ($[-2.25, 0.28]$ mm), all relative to FullVirtual. A focused recurrent-history isolation changed TCP RMSE by −0.67 mm with an interval crossing zero. Thus the evidence supports temporal input consistency but does not establish an independent nominal tracking gain from recurrent-history advancement.

Projection, reranking, and effort sensitivity. The detailed projection and task-score studies are in `abb/summaries/projection_choice_aggregate.csv` and `abb/summaries/task_cost_aggregate.csv`. At common $D = 6$, task-space reranking did not establish an independent tracking gain. The post-freeze sweep jointly scales servo and residual-smoothness terms by $0.5, 1, 2, 4, 8\times$; the complete tracking, acceleration, torque RMS/P95/max, jerk, and power proxies are in `claim_evidence/summaries/effort_pareto_aggregate.csv` and `claim_evidence/summaries/effort_peak_aggregate.csv`. It is a diagnostic frontier, not a replacement primary configuration.